MUKUL SHARMA

Senior Staff Engineer · Software factories for AI coding agents · Ships 1.0 products

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I take products from first commit to version 1.0. In 16 years I shipped SOASTA TouchTest, the mPulse Mobile SDK, Speak Music's Muse, MobileIron AppStation, Adobe Express video, and the Adobe Firefly Video Editor timeline, which reached 100,000 monthly users in public beta. Now I build software factories where AI coding agents write production code. Architect, my agent system, wrote the code for 400+ of my merged pull requests in 6 weeks.

EXPERIENCE

Senior Computer Scientist 2 (Senior Staff Engineer), Adobe, San Jose, CA *Aug 2021 – Present*

- Senior engineer for my team of 22 software and quality engineers for 3.5 years. Reviewed about 2,300 pull requests and mentored 10 engineers.

Software factory *2026*

- Built Architect, an agent system that turns a problem statement into a reviewed pull request that passes CI. Competing plans face critics, agents write and mutation-test the code, and a person approves. 16+ engineers have run it about 2,000 times.
- Merged 400+ pull requests into Adobe Firefly's platform in 6 weeks, with production code written by Architect. 787 merged since April 2026.
- Wrote evals that pin each agent's behavior. Each eval fails before a change and passes after it, and CI blocks a prompt edit that breaks one. A coverage policy sets the test layer each change needs.
- Wrote the Claude Code skills, rules, and review checks that guide agents through Adobe's video and Firefly code. Trained engineers to use them and helped the Lightroom and Firefly Boards teams adopt them.
- Open-sourced Forge (competing agent plans, critics, and a judge, then a step-by-step pipeline), ProofGate (checks agents' claims against the commands they ran; rules from 94,000 real commands), Assistant (starts and tracks my Claude Code sessions; 500+ since May 2026), and Perch (Claude Code sessions from Slack).

Adobe Firefly Video Editor and Firefly video Assembler *2023 – Present*

- Built the timeline editing core from the first commit: multi-track and freeform tracks, trim, move, copy and paste, and keyboard shortcuts. The public beta launched in Dec 2025 and reached 100,000 monthly active users.
- Led cross-team work on WebCodecs audio, audio-video sync, and frame decoding that cleared public beta blockers. Stabilized the import and first-edit flows.
- Contributed the time coordinate system for frame-accurate editing and frame-rate support that matches Premiere Pro. Helped build caching that made Move 70% faster on a 10-track timeline.
- Leading delivery of Firefly video Assembler to public beta in November 2026.

Adobe Express and Project Violet (iOS) *2021 – 2023*

- Led the team that built 8 Video Quick Actions for the Adobe Express launch, and fixed several launch blockers.
- Led the team that delivered the first video timeline in Adobe Express for iPhone and iPad (Mobile Sceneline v1).
- Built Dynamic Text, log export, a Project Editor refactor for parallel animations, and the Soundtrack Selector for Project Violet, a SwiftUI video app.

Staff Software Engineer (iOS), MobileIron (acquired by Ivanti), Mountain View, CA *Oct 2018 – Aug 2021*

- Built and shipped AppStation 1.0 for iOS (Jun 2019), MobileIron's app for managing work apps without managing the whole phone.
- Lead engineer for MobileIron's enterprise iOS apps, including Mobile@Work and MobileIron Go, in Swift and Objective-C. Owned the release process and app container security.

iOS Platform Lead, Speak Music, San Jose, CA *Jan 2017 – Oct 2018*

- Led design, development, and launch of Muse, an Alexa device for the car, in Nov 2017. Muse raised $70,000+ from 1,000+ Indiegogo backers. Owned the Muse iOS app from start to delivery.

- Took over the Melody iOS app, a voice music assistant for iHome speakers, on joining.

Senior Mobile Engineer, 8x8, San Jose, CA *Dec 2015 – Jan 2017*

- Lead developer for the 8x8 Virtual Office iOS app. Set up its iOS and Android test automation with TouchTest and Jenkins.

Senior Software Engineer, then Tech Lead, SOASTA, Mountain View, CA *May 2011 – Dec 2015*

- On the launch team for SOASTA TouchTest (2012), mobile test automation for iOS and Android.
- Designed and built the mPulse Mobile SDK, its API, the communication model, and network-request analytics.
- Built tooling that injects SOASTA libraries into iOS and Android app files (IPA and APK), cutting customers' time to first test. Promoted to Tech Lead in 2015.

Software Engineer, SumTotal Systems, Gainesville, FL *Jun 2010 – May 2011*

- Led OpenScript development for version 11 of the ToolBook authoring suite.

SKILLS

AI and agents Multi-agent harnesses, evals and LLM judges, MCP, tool calling, Claude Code, RAG

Languages TypeScript, JavaScript, Python, Swift, Objective-C, Java

Platforms React, Node.js, iOS, Android, WebCodecs, Playwright, Jenkins, Git

PATENTS AND PUBLICATIONS

Load test charts with standard deviation and percentile statistics · *US Patent 9,251,035, issued Feb 2016*

Cloud-based custom metric/timer definitions and real-time analytics of mobile apps · *US 2015/0067527, published Mar 2015*

SHIELD: Social Sensing and Help in Emergency using Mobile Devices · *G. S. Thakur, M. Sharma, A. Helmy, IEEE Globecom 2010*

EDUCATION

M.S. in Computer Science, University of Florida (GPA 3.90 / 4.00) *May 2010*